Discussion Exercise 3

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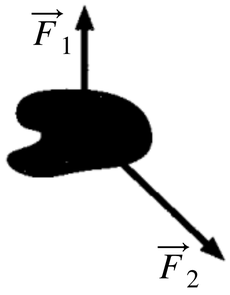
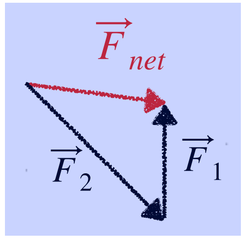
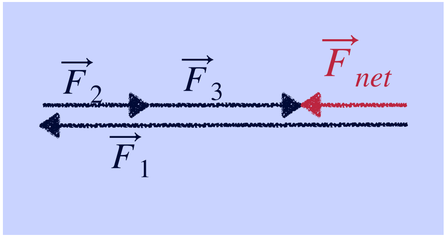
1. Net Force

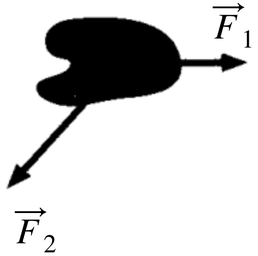
1. Two or more forces are shown on the objects below. Draw and label the net force $\vec{F}_{net}$.

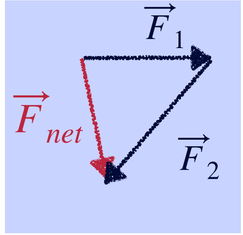


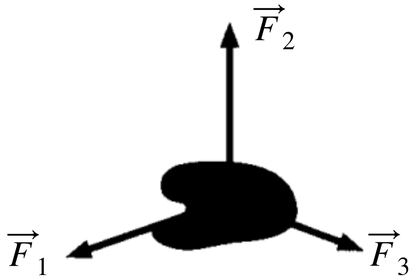


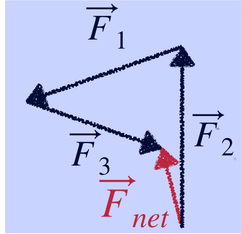


2. Two or more forces are shown on the objects below. Draw and label the net force $\vec{F}_{net}$.









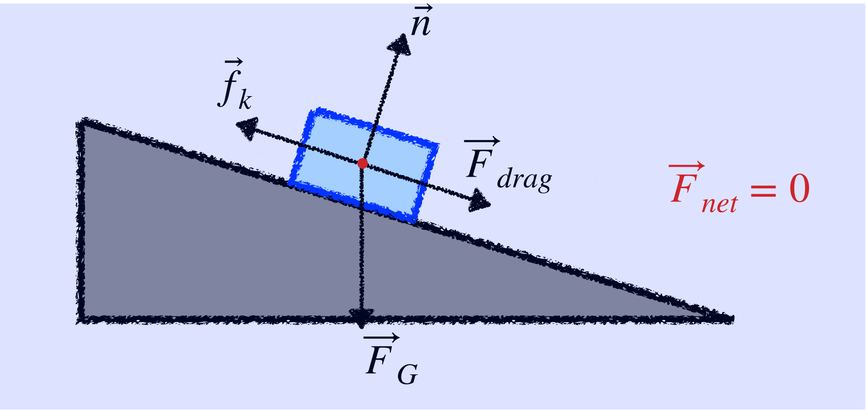
2. Free-Body Diagrams

Exercises 24–29:

- Draw a picture and identify the forces, then
- Draw a complete free-body diagram for the object, following each of the steps given in Tactics Box 4.3. Be sure to think carefully about the direction of $\vec{F}_{net}$.

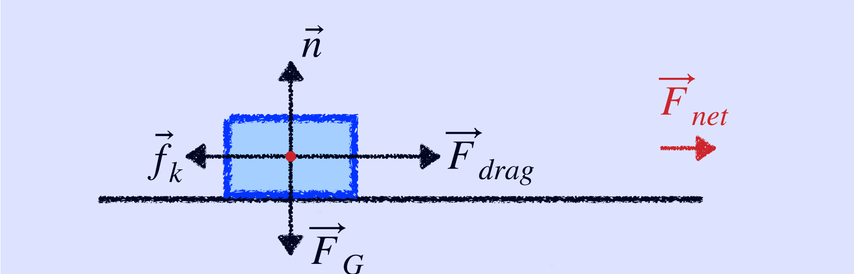
Note: Draw individual force vectors with a **black** or **blue** pencil or pen. Draw the *net* force vector $\vec{F}_{net}$ with a **red** pencil or pen.

24. A heavy crate is being lowered straight down at a constant speed by a steel cable.



$\vec{F}_{net} = 0$ *since it moves with a constant velocity*

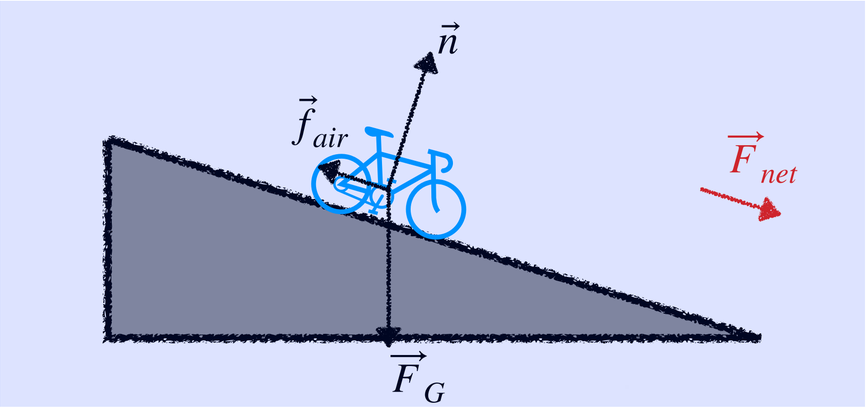
25. A boy is pushing a box across the floor at a steadily increasing speed. Let the box be "the system" for analysis.



$\vec{F}_{net}$ *is nonzero since the box moves with a finite acceleration*

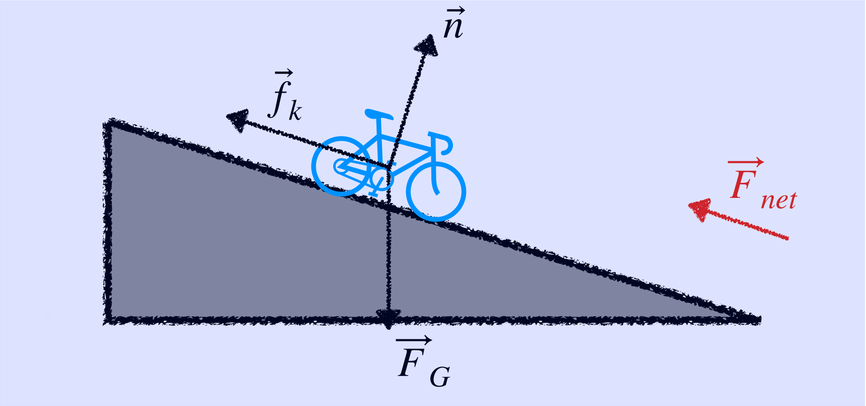
2. Free-Body Diagrams

26. A bicycle is speeding up down a hill. Friction is negligible, but air resistance is not.



$\vec{F}_{net}$ points downward along the hill

27. You've slammed on your car brakes while going down a hill. You're skidding to a halt.

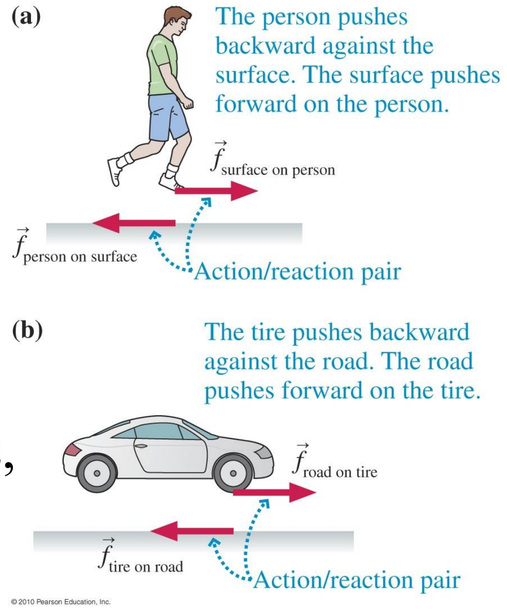


$\vec{F}_{net}$ points upward along the hill

The direction of the net force should be the same as that of the acceleration due to $\vec{F}_{net} = m\vec{a}$

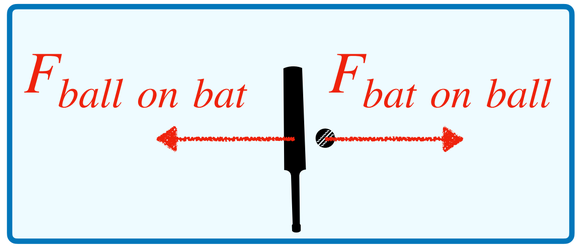
3. Newton's Third Law

Each of the following situations has two or more interacting objects. Draw a picture similar to the figure on the right (Fig 4.31 in the textbook), in which you show

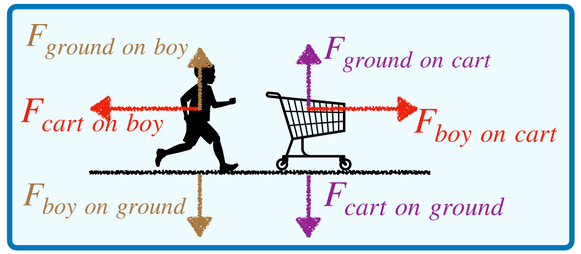


- Show the interacting objects, with a small gap between them.
- Draw the force vectors of all action/reaction pairs.
- Label the force vectors, using a notation like $\vec{F}_{AonB}$ or $\vec{F}_{BonA}$.

1. A bat hits a ball. Draw your picture from the perspective of someone seeing the end of the bat at the moment it strikes the ball. The objects are the bat and the ball.

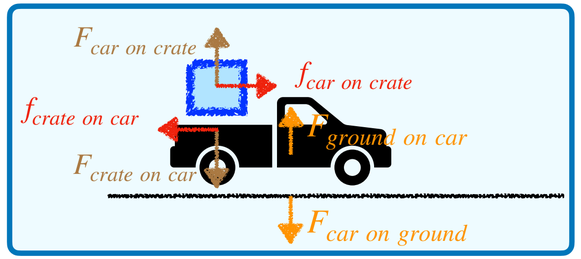


2. A boy pulls a wagon by its handle. Rolling friction is not negligible. The objects are the boy, the wagon and the ground.



Don't forget the action/reaction pairs between the objects and the ground!

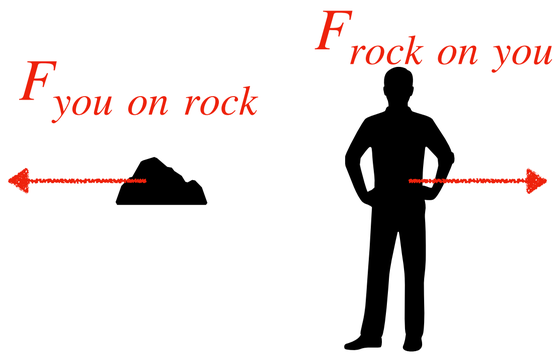
3. A crate is in the back of a truck as the truck accelerates forward. The crate does not slip. The objects are the truck, the crate and the ground.



f is due to the friction between the car and the crate

4. Exercise 4

You find yourself in the middle of a frozen lake with a surface so slippery that you cannot walk. However, you happen to have several rocks in your pocket. The ice is extremely hard. It cannot be chipped, and the rocks slip on it just as much as your feet do. Can you think of a way to get to shore? Use pictures, forces and Newton's laws to explain your reasoning.

If you hurl the rocks in the opposite direction of where you want to go, you will be pushed in the right way by the rocks.

*By **Newton's third law**, one exerts a force in one direction on the rock, and the rock exerts an equal force in the opposite direction on that person. This force will lead that person to slide across the ice in the opposite direction in which the rock was thrown.*